

# AI Culture Hub

## Data Modeling & Architecture Documentation

### 1. Introduction & Project Overview

AI Culture Hub is a comprehensive Learning Management System (LMS) designed for AI and Digital Transformation education within organizations. The platform enables employee training on AI concepts, tools, and best practices through a modern, gamified learning experience.

#### 1.1 Key Features

- Course and lesson management with enrollment tracking
- Quiz-based assessments with automatic scoring
- Gamification system with points, levels, badges, and leaderboards
- Role-based access control (RBAC) with 51 granular permissions
- Article management with categories and tags
- User progress tracking and learning streaks

#### 1.2 Technology Stack

Frontend: Angular 17 with standalone components, Bootstrap 5, RxJS

Backend: ASP.NET Core 8.0 Web API with Entity Framework Core 8.0

Database: Microsoft SQL Server with code-first migrations

Authentication: JWT Bearer Tokens with 60-minute expiration

#### 1.3 Project Architecture

The solution follows Clean Architecture with distinct layers:

API Layer: Controllers handling HTTP requests with Swagger documentation

Application Layer: DTOs, Interfaces, and business logic abstractions

Domain Layer: Entity definitions representing core business concepts

Infrastructure Layer: Database context, service implementations

## 2. Data Model Overview

The database schema contains 28 tables organized into six logical domains. All tables include standard audit fields and soft-delete support through IsActive flags.

### 2.1 Entity Relationship Summary

- CORE: Users, Roles, Permissions, UserRoles, RolePermissions
- CONTENT: Articles, Categories, Tags, ArticleTags, ArticleViews, Bookmarks
- LEARNING: Courses, Lessons, CourseEnrollments, UserProgress
- ASSESSMENT: Quizzes, Questions, Answers, QuizAttempts, QuizAttemptAnswers
- GAMIFICATION: Levels, Badges, UserLevels, UserBadges, PointTransactions, Challenges
- SYSTEM: Notifications, Announcements, Glossary, AuditLogs, SystemSettings, Themes

### 2.2 Core User & Security Model

The security model implements many-to-many relationships between Users and Roles, and between Roles and Permissions. This flexible design allows for:

- Multiple roles per user (e.g., Editor who is also Instructor)
- Granular permission assignment at the role level
- Consistent permission checking through [RequirePermission] attribute

Users table tracks gamification data (TotalPoints, CurrentLevelId, LearningStreak) and preferences (Themeld, LastActivityDate).

## 3. Database Schema Details

### 3.1 Users Table

Stores user account information:

- Username, Email (both unique indexed)
- PasswordHash, PasswordSalt for secure authentication
- TotalPoints, CurrentLevelId, CurrentLevelPoints for gamification
- LearningStreak, LastActivityDate for engagement tracking
- ThemeId for UI preference, AvatarUrl for profile

### 3.2 Roles & Permissions

Roles: Administrator, ContentManager, Editor, Instructor, Employee

Permissions: 51 total, organized across 12 modules (Dashboard, Users, Roles, Articles, Categories, Courses, Quizzes, Challenges, Announcements, Reports, Settings, Glossary)

### 3.3 Courses & Lessons

Courses: Title, Description, ThumbnailUrl, Difficulty

Engagement: EnrolledCount, CompletionCount, AverageRating

Lessons: Belongs to Course (CASCADE delete), Contains Content, VideoUrl, OrderIndex

### 3.4 Quizzes & Assessment

Question types: MultipleChoice, TrueFalse, MultipleAnswer

QuizAttempts: Score, MaxScore, Percentage, CorrectAnswers, TotalQuestions

QuizAttemptAnswers tracks individual answer records for audit.

## 4. Learning & Gamification Entities

### 4.1 Progress Tracking

CourseEnrollments: ProgressPercentage (decimal), Status (Enrolled/InProgress/Completed)

UserProgress: Links Users to Lessons, IsCompleted flag, CompletionDate, TimeSpentMinutes

### 4.2 Levels & Progression

5 tiers with PointsRequired thresholds:

AI Curious (0) -> AI Explorer (500) -> AI Enthusiast (1000) ->

AI Expert (2000) -> AI Champion (3000)

UserLevels maintains achieved levels per user with AchievedDate.

### 4.3 Badges & Achievements

Flexible criteria: CriteriaType (courses\_completed, quizzes\_passed, etc.), CriteriaValue threshold

Unique constraint on (UserId, BadgeId) prevents duplicates.

### 4.4 Daily Challenges

ChallengeType: Daily, Weekly, Special

QuestionText, Options (JSON), CorrectAnswer, Points, Difficulty, ActiveDate

### 4.5 Point Transactions

TransactionType: ArticleRead, CourseComplete, QuizPassed, DailyLogin

ReferenceType/Id for linking to source entity.

Indexed by UserId and TransactionDate.

## 5. System Architecture & Security

### 5.1 API Architecture

RESTful API with 11 controllers:

AuthController: Login, Register, Profile, Password changes

AdminController: User management, Roles, System configuration

ArticlesController: CRUD with publishing workflow

CoursesController: Enrollment, Lesson completion

QuizzesController: Assessment creation, Submission, Scoring

ChallengesController: Daily challenges

DashboardController: Aggregated user statistics

### 5.2 Authentication & Authorization

JWT: 60 minutes expiry, Secret via JWT\_SECRET environment variable

Custom RequirePermissionAttribute for granular access control

Password hashing with salt, Audit logging, Unique constraints

### 5.3 Frontend Architecture

Angular 17 standalone components with lazy-loaded features:

Core (singletons): Services, Guards, Interceptors

Features: auth, dashboard, articles, courses, quizzes, leaderboard, admin

AuthInterceptor injects JWT Bearer token. AuthGuard protects routes.

### 5.4 Database Configuration

Connection: Server=localhost;Database=AlCultureHub

Entity Framework Core 8.0 handles migrations

NVARCHAR for Unicode support (Persian department names)

### 5.5 Key Design Patterns

Repository Pattern: Data access through services

DTO Pattern: Separate objects for API requests/responses

Custom Attributes: RequirePermission for authorization

Index Optimization: Composite indexes on frequent queries

Soft Deletes: IsActive flags preserve data integrity